

Aggregate Flexibility of Virtual Power Plants With Temporal Coupling Constraints

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Abstract—The virtual power plant (VPP) provides an efficient way to manage a bunch of distributed energy resources (DERs). In order to participate in the system operation and power market as a whole, the estimation of VPP power flexibility is indispensable. However, the temporal coupling constraints make it intractable to aggregate the power flexibility directly. In this paper, we formulate the flexibility evaluation problem as a chance constrained optimization model incorporated the randomness of renewables. To make it tractable, the evaluated flexibility is represented by the combination of a virtual generator and a virtual battery which is decomposed by a robust optimization algorithm. Then, a novel high-dimensional polytope based bound shrinking method is proposed to estimate the parameters of the virtual generator and virtual battery. The numerical test results show the proposed method obtain much more precise results with lower conservativeness than those of existing methods. The disaggregation feasibility of the evaluation results is also verified.

Index Terms—Virtual power plant, aggregate power flexibility, high-dimensional polytope projection.

NOMENCLATURE

Variables

$\mathbf{x}$	Vector of decision variables, consisting of the active and reactive power of all the DERs at all time periods
$\mathbf{x}_t$	Vector of decision variables at time t
$\mathbf{x}_t^G$	Active power of generator-like DERs
$\mathbf{x}_t^B$	Active power of battery-like DERs
$\mathbf{x}_t^Q$	Reactive power of all DERs
$\tilde{\mathbf{e}}$	Vector of forecast errors
$\mathbf{p}^{\text{VPP}}$	Output active power trajectory of VPP
p_t^{VPP}	Output active power of VPP at time t
q_t^{VPP}	Output reactive power of VPP at time t
$\mathbf{p}^G, \mathbf{p}^B$	Output active power trajectory of VG and VB
$\mathbf{b}_{(k)}^G$	Vector of VG's key parameters in the k -th iteration
$\mathbf{p}_{(k)}^G$	Infeasible extreme points of $\Omega_{(k)}^G$
$\mathbf{p}_{(k)}^{\text{bd}}$	Feasible bound points found in the k -th iteration.

Notation

$\mathcal{T}$	Discrete-time horizon set
T	Total number of discrete-time horizon
$\mathcal{R}_t$	Active-reactive power flexibility range at time t
Ω_1^G	Origin flexibility region of GDERs
Ω_2^G	Precise projected flexibility region of VG
$\Omega_{(k)}^G$	Flexibility region of VG obtained in the k -th iteration
Ω^G	Aggregated flexibility region of VG
Ω^B	Aggregated flexibility region of VB
Ω^{VPP}	Flexibility region of VPP.

I. INTRODUCTION

A. Motivation

THE NUMEROUS distributed energy resources (DERs) in the distribution network, such as combined heat and power generators (CHPs), energy storage systems (ESSs), photovoltaics (PVs), wind turbines (WTs), heating, ventilation, and air conditioning systems (HVACs) have great potential of flexibility. However, the DERs are small-scale, distributed and heterogeneous, which makes it very difficult to manage them and utilize their power flexibility. The virtual power plant (VPP) provides an efficient way to manage the DERs in a cluster way [1]. The operator of a VPP can aggregate flexibility of DERs as a whole to provide energy and auxiliary service to the grid. Therefore, evaluating the equivalent power flexibility of the VPP by aggregating DERs is significantly important. It can provide scientific decision-making basis for dispatching in power system operation and bidding in the power market.

The power flexibility can be described as a feasible region of active and reactive output power over all time periods, which is related to the technical operational constraints. In this paper, we focus on the trajectory of active and reactive output power at the point of common coupling (PCC), so that the aggregated power flexibility of VPP is the feasible region of the trajectories. From the perspective of geometric, the flexibility aggregation process can be regarded as the projection from the power flexibility region of all the DERs to the power flexibility region of VPP spanned by the output power at PCC [2]–[5]. Therefore, this problem can be described as a projection from a high-dimensional origin polytope to a reduced dimensional projected polytope. It is mathematically an intractable problem to obtain the precise analytical form of the projected polytope.

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In this paper, we propose an efficient method to evaluate the equivalent power flexibility of VPP incorporating the temporal coupling operational constraints of DERs and the network constraints as well as their randomness. The core idea is to equivalently formulate the VPP as a virtual generator and a virtual battery.

B. Literature Review

Estimating the flexibility of aggregated DERs has been studied in several works. Some works presented the methods to estimate the aggregated active-reactive power flexibility range in a single period, such as [6]–[10]. Hao *et al.* propose a method to aggregate a collection of homogeneous adjustable loads in a smart community, and use virtual battery model to represent the flexibility in [11], [12]. The work of [13] aggregates the flexibility of heat pumps based on the Minkowski Sum method, and the cluster of heat pump is equivalent to a virtual synchronous machine or a virtual energy storage. Reference [3] uses polytopic projection approximation method to extract the flexibility of heterogeneous deferrable loads. In [14], the flexibility of each individual load is approximated with a homothetic polytope, then the flexibility of the whole cluster can be calculated by Minkowski Sum easily. Reference [15] presents a method to approximate the tie-line security region of a regional network with the ramp rates of generators. All the works mentioned above focus on the power flexibility of similar types of DERs, and some of them do not consider the network constraints among the DERs.

There are also some other works, which tries to describe the aggregated power flexibility of DERs in the special forms. In reference [16], the flexibility of VPP is equivalent to a generator-like model with the upper and lower power bounds, ramp rates, and energy limits as the aggregated parameters. Chen *et al.* propose a distribution-level power aggregation method to approximate the feasible region of a multi-phase distribution network [5], in which the upper and lower bounds of output power at each time period were obtained. Then, they improve the solving method with the two-stage adaptive robust optimization algorithm in [17]. In [18], the flexibility region is accessed by the ellipsoidal inner approximation method and further improves the evaluation performance. With these methods, the flexibility of aggregated DERs is efficiently evaluated, the results, however, are still very conservative. This is because that the flexibility of heterogeneous DERs are mixed together, and the parameters of aggregated flexibility region has to compromise to guarantee the disaggregation feasibility. Therefore, the complementary characteristic of active power in different time periods cannot be fully exploited.

C. Contribution and Paper Organization

In this paper, we propose a novel method to evaluate the aggregated power flexibility of VPP. Different types of DERs and network constraints are taken into consideration.

Firstly, the Gaussian mixture model (GMM) is used to fit the joint probability density function of DERs' output and the operational constraints are formulated as chance constraints. Since the linear combination of GMM variables is

still a GMM variable [19], the stochastic constraints can be transferred into the deterministic ones [20]. The heterogeneous DERs in a VPP can be categorized into two groups of homogeneous DERs according to their different operational constraints. One is the generator-like DERs and the other is the battery-like DERs. Furthermore, the generator-like DERs can be aggregated as a virtual generator (VG), and the battery-like DERs can be aggregated as a virtual battery (VB). Finally, the high-dimensional polytope based bound shrinking method is proposed to calculate the parameters of VG and VB. This method not only guarantees the disaggregation feasibility of the evaluation result, but also fully exploit the flexibility of DERs.

The main contributions of this paper are threefold:

1) The temporal coupling constraints of DERs are taken into consideration as well as the network constraints, which involves high-dimensional interdependency flexibility region. In this paper, the evaluation of aggregated power flexibility is modeled as a projection problem of high-dimensional polytope. The aggregated power flexibility region of VPP is approximated by a reduced dimensional polytope with a specific shape.

2) A VG and a VB are used to represent the aggregated power flexibility of VPP, which has a clear physical meaning and much lower conservatism compared with the existing methods.

3) A novel polytope based bound shrinking method is proposed to calculate the parameters of the equivalent VG and VB, which not only explores the flexibility but also guarantee the disaggregation feasibility.

The rest of the paper is organized as follows. The process of power flexibility aggregation method is described in Section II. Section III introduces the method to obtain the parameters of VG and VB. The numerical tests are carried out to demonstrate the performance of our proposed method in Section IV and the conclusions are drawn in Section V.

II. POWER FLEXIBILITY AGGREGATION METHOD

A. Overview of Aggregation Method

The flowchart of the proposed power flexibility aggregation method is shown in Fig. 1. The inputs of our method includes the network model, DERs model and the historical data of random variables. Their details are presented in Section II-B.

There are some chance constrained inequalities involving the random variables in the model. Section II-C introduces the method to transform them to the equivalent deterministic constraints. Then, the stochastic programming problem becomes a deterministic one.

Due to the energy constraints, thermal constraints, or ramp rate of DERs, the active power at different time periods are interdependent, while the reactive power at different time periods is decoupled. Accordingly, the power flexibility aggregation problem can be separated into two stages: (i) evaluate the temporal decoupling active-reactive power flexibility ranges at each time periods, and (ii) estimate the temporal coupling flexibility of active power trajectory at all time periods.

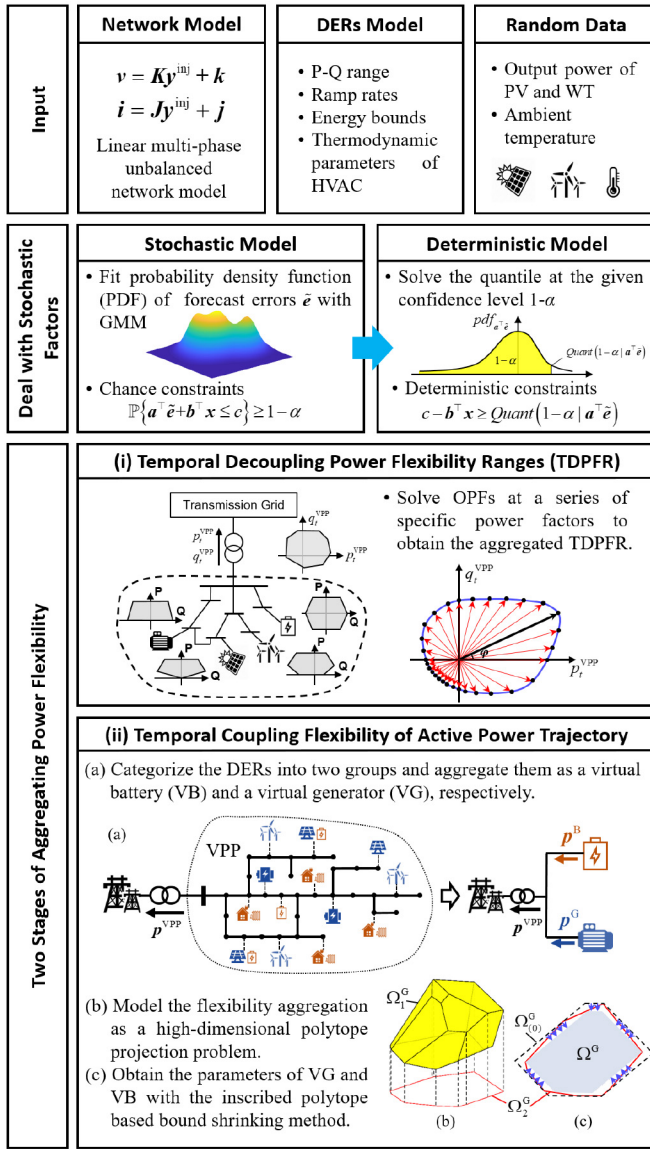


Fig. 1. Flowchart of aggregating the power flexibility of VPP.

For the first stage, the power capability ranges of DERs are aggregated as the active-reactive power flexibility range at PCC of VPP at each time period. The temporal coupling constraints are omitted, while the operational constraints at single time period, such as the network constraints, power capability charts of DERs and forecast maximum output power of renewable generators, are taken into consideration in this process. The solution process of this part is shown in Section II-D.

The second stage is the core of this paper. All the constraints including temporal coupling ones are taken into consideration. According to the constraints types of all the DERs in the VPP, they can be divided into two groups of homogeneous DERs. One group contains the generator-like DERs (GDERs), including the CHP, PV, WT, and uncontrollable loads. These DERs have the constraints of upper and lower bounds of output active power, and some of them have ramp rates. The other group contains the battery-like DERs (BDERs), including ESS and HVAC. These DERs have the temporal coupling

energy or temperature constraints. Then, the former group of DERs is aggregated to a VG, and the latter one is aggregated to a VB. Finally, the flexibility of VPP can be regarded as the sum of the flexibility of these two virtual components, as shown in (a) of part (ii) of Fig. 1. Then, a bound shrinking method is proposed to obtain their parameters, respectively. Section II-E presents the detailed solution process.

B. Input Data and Models

In order to estimate the power flexibility of VPP, the network model, the DERs model, and historical random data are needed as the input information.

Besides, we use the linear multi-phase unbalanced network model [21], [22] to calculate the power flow of VPPs. The magnitudes of buses' voltage v , branches' current i and injection power at PCC p^0 and q^0 can be expressed as the following affine form:

$$v = Ky^{\text{inj}} + k \quad (1)$$

$$i = Jy^{\text{inj}} + j \quad (2)$$

$$p^0 = g^T y^{\text{inj}} + d \quad (3)$$

$$q^0 = h^T y^{\text{inj}} + l \quad (4)$$

where y^{inj} is a vector composed of the injection active and reactive power of all phases of the buses. K , J , g , h , k , j , d , and l are the system parameters and the methods to obtain them are shown in [21], [22]. With this network model, the network constraints can be expressed as linear constraints.

The DERs' models consist of two types of technical constraints of DERs, the temporal decoupling constraints and temporal coupling constraints. The former includes the power capability of DERs and the maximum output power of renewable generators, as presented in [23]. The latter includes the ramp rates of CHPs, energy constraints of ESSs and temperature constraints of HVACs [5]. To simplify the optimization problem and accelerate the computation process, all the DERs' models are approximately linearized. Even though the start-up/shutdown actions of CHPs and charging/ discharging states of ESSs may introduce binary variables, they can still be linearized. On one hand, the start-up/shutdown of CHP generators is omitted, because these generators in the distribution networks are small-scale and have fast response speed. On the other hand, the conflicts of charging and discharging can be avoided without binary variables [24]. By adding a penalty term for charging and discharging loss, the nonlinear complementary constraint of ESSs can be relaxed.

The random variables in this paper include the ambient temperature, the maximum output power of PV and WT, and the uncontrollable loads. The input historical random data are used to generate the joint probability density functions of their forecast error.

C. Deal With Stochastic Factors

Due to the stochastic factors in the VPP model, the constraints involving the random variables can be expressed as the chance constraints, which means the constraints hold at a specific confidence level. A general form of chance

constraint is:

$$\mathbb{P}\left\{\mathbf{a}^\top \tilde{\mathbf{e}} + \mathbf{b}^\top \mathbf{x} \leq c\right\} \geq 1 - \alpha \quad (5)$$

where $\mathbf{x}$ denotes the vector of decision variables, which consists of the active and reactive power of all the DERs at all time periods; $\tilde{\mathbf{e}}$ denotes the vector of forecast errors; $\mathbf{a}$, $\mathbf{b}$, and c are constant parameters; and $1 - \alpha$ is the specific confidence level. With the Gaussian mixture model (GMM), the joint probability function of forecast errors $\tilde{\mathbf{e}}$ can be fitted with the linear combination of several Gaussian functions. Then the quantile of the random variable $\mathbf{a}^\top \tilde{\mathbf{e}}$ at confidence level $1 - \alpha$ can be calculated. The chance constraint (5) can be transformed into an equivalent constraint as follows [19].

$$c - \mathbf{b}^\top \mathbf{x} \geq \text{Quant}\left(1 - \alpha | \mathbf{a}^\top \tilde{\mathbf{e}}\right) \quad (6)$$

where $\text{Quant}(1 - \alpha | \mathbf{a}^\top \tilde{\mathbf{e}})$ denotes the quantile of the random variable $\mathbf{a}^\top \tilde{\mathbf{e}}$ at confidence level $1 - \alpha$.

In this way, all the stochastic constraints can be transformed into equivalent deterministic ones. Accordingly, the stochastic problem is transformed into a deterministic one.

By incorporating the linear network constraints and DER models, the technical constraints of VPP can be obtained. To simplify the expression, all the constraints can be organized into a compact form as follows:

$$\mathbf{M}_t \mathbf{x}_t \leq \mathbf{m}_t, \forall t \in \mathcal{T} \quad (7)$$

$$\mathbf{E} \mathbf{x} \leq \mathbf{f} \quad (8)$$

$$\mathbf{p}^{\text{VPP}} = \mathbf{C} \mathbf{x} + \mathbf{d} \quad (9)$$

where $\mathbf{x}_t$ denotes the vector of decision variables at time t , consisting of the DERs' active and reactive power at time t . Constraint (7) is the collection of DERs' temporal decoupling constraints and network constraints at time t . Constraint (8) is the collection of all the DERs' constraints and network constraints. The output active power trajectory of VPP is expressed as (9).

D. Temporal Decoupling Active-Reactive Power Flexibility Ranges at Each Time Period

If omitting the temporal coupling constraints, the active-reactive power flexibility at a specific time period can be expressed by a range with a specific shape in the P-Q pane. In the work [4], [6], [10], [23], the contour of the active-reactive power flexibility ranges at multiple time periods are calculated by solving a series of optimal power flow (OPF) problems and then simplified with the piecewise convex linear approximation method. The active-reactive power flexibility ranges as each time period can be expressed as the following polygon form:

$$\mathcal{R}_t = \left\{ (p_t^{\text{VPP}}, q_t^{\text{VPP}}) \left| \mathbf{A}_t^{\text{VPP}} \begin{bmatrix} p_t^{\text{VPP}} \\ q_t^{\text{VPP}} \end{bmatrix} \leq \mathbf{b}_t^{\text{VPP}} \right. \right\}, \forall t \in \mathcal{T} \quad (10)$$

where $\mathbf{A}_t^{\text{VPP}}$ and $\mathbf{b}_t^{\text{VPP}}$ denote the parameters of the active-reactive power flexibility range at time t .

E. Temporal Coupling Flexibility of Active Power Trajectory at All Time Periods

1) *Difficulties of Temporal Coupling Flexibility Aggregation*: The aggregation of temporal coupling flexibility problem can be regarded as a projection problem from a high-dimensional original polytope Ω_1 to a projected polytope Ω_2 . Ω_1 and Ω_2 can be described as

$$\Omega_1 = \left\{ \begin{bmatrix} \mathbf{x} \\ \mathbf{p}^{\text{VPP}} \end{bmatrix} \left| \begin{array}{l} \mathbf{E} \mathbf{x} \leq \mathbf{f}, \\ \mathbf{p}^{\text{VPP}} = \mathbf{C} \mathbf{x} + \mathbf{d} \end{array} \right. \right\} \quad (11)$$

$$\Omega_2 = \{ \mathbf{p}^{\text{VPP}} | \mathbf{\Pi}^* \mathbf{p}^{\text{VPP}} \leq \mathbf{\pi}^* \} \quad (12)$$

where the original polytope Ω_1 describes the feasible operational region of all the DERs. It consists of the operational constraints of all the DERs, the network constraints, and the affine equation between the output power of DERs and output power at PCC of VPP. The projected polytope Ω_2 describes the power flexibility of VPP. $\mathbf{\Pi}^*$ and $\mathbf{\pi}^*$ denote the parameters of the aggregated flexibility region.

Unfortunately, the accurate values of $\mathbf{\Pi}^*$ and $\mathbf{\pi}^*$ cannot be obtained easily, because the dimension of the projected polytope Ω_2 is still high, and the number of hyperplanes forming Ω_2 is extremely large. Actually, the general high-dimensional projection problem is an NP-hard problem. Therefore, we solve this problem by leveraging the special characteristics of homogeneous DERs. The flexibility aggregation of homogeneous DERs is intuitively simple, because they have the same types of technical constraints. Furthermore, the cluster composed of a group of homogeneous DERs has similar technical constraints as an individual DER, too. For example, consider a cluster composed of several CHPs. Each CHP has the upper and lower bounds of output power, and the ramp rates as their flexibility parameters. Then, the aggregated cluster can be regarded as a virtual CHP with the power bounds and ramp rates as the technical constraints.

2) *Separate the Network Constraints of Two Types of DERs*: Due to the network constraints among the DERs, the power of GDERs and BDERs are namely coupled together. In this section, the network constraints of two types of DERs are separated. Then, the network constraints of GDERs can be always satisfied regardless of the active power of BDERs, and vice versa. The robust optimization algorithm can be used to eliminate the coupling constraints between GDERs and BDERs.

For the simplicity of the formula, we divide $\mathbf{x}_t$ into three parts, which are the active power of GDERs $\mathbf{x}_t^{\text{G}}$, the active power of BDERs $\mathbf{x}_t^{\text{B}}$, and the reactive power of all DERs $\mathbf{x}_t^{\text{Q}}$, respectively. Then, the temporal decoupling constraints (7) at time t can be expressed as:

$$\mathbf{M}_t \left[\begin{pmatrix} \mathbf{x}_t^{\text{G}} \\ \mathbf{x}_t^{\text{B}} \\ \mathbf{x}_t^{\text{Q}} \end{pmatrix}^\top \right]^\top \leq \mathbf{m}_t \quad (13)$$

We first regard $\mathbf{x}_t^{\text{B}}$ as the uncertain parameters in the robust optimization problem. The constraints only relevant to $\mathbf{x}_t^{\text{B}}$ can be separated, so that (13) can be further rearranged as

$$\begin{bmatrix} \mathbf{0} & \mathbf{M}_t^{\text{G}} & \mathbf{0} \\ \mathbf{M}_t^{\text{B}} & \mathbf{M}_t^{\text{B}} & \mathbf{M}_t^{\text{Q}} \end{bmatrix} \begin{bmatrix} \mathbf{x}_t^{\text{G}} \\ \mathbf{x}_t^{\text{B}} \\ \mathbf{x}_t^{\text{Q}} \end{bmatrix} \leq \begin{bmatrix} \mathbf{m}_t^{\text{G}} \\ \mathbf{m}_t^{\text{B}} \\ \mathbf{m}_t^{\text{Q}} \end{bmatrix} \quad (14)$$

Then, the robust optimization problem can be expressed as follows: For any given x_t^B that meets

$$M_t^0 x_t^B \leq m_t^0 \quad (15)$$

the following inequality always holds

$$M_t^G x_t^G + M_t^B x_t^B + M_t^Q x_t^Q \leq m_t^1 \quad (16)$$

With the robust counterpart method, the robust optimization problem can be equivalently transformed into a series of linear programming problems.

$$\left(M_t^G x_t^G + M_t^Q x_t^Q \right)_i \leq \min_{x_t^B} \left\{ \left(m_t^1 - M_t^B x_t^B \right)_i \mid M_t^0 x_t^B \leq m_t^0 \right\} \quad (17)$$

where $(\bullet)_i$ denotes the i -th row of a matrix or the i -th element of a vector. Finally, the decoupled network constraints of GDERs at each time period can be expressed as:

$$N_t^G x_t^{GQ} \leq n_t^G \quad (18)$$

where $x_t^{GQ} := [(x_t^G)^\top, (x_t^Q)^\top]^\top$.

With similar method, we can obtain the decoupled network constraints of BDERs at each time period as follows:

$$N_t^B x_t^{BQ} \leq n_t^B \quad (19)$$

where $x_t^{BQ} := [(x_t^B)^\top, (x_t^Q)^\top]^\top$.

Discussion: The robust optimization method always introduce conservation to the result, since the network constraints must be guaranteed even in the worst scenarios. However, there is usually sufficient transmission capacity inside VPP and voltage regulation facilities in distribution networks. The network congestion will seldom occur in the VPP. Therefore, the separation of these two types of DERs will not make our method over conservative.

3) *Flexibility of VG and VB:* The decoupled network constraints of two groups of homogeneous DERs are obtained as (18), (19). According to the analyses above, these two types of homogeneous DERs can be further aggregated as a VG and a VB, respectively. In this section, we propose a polytope based bound shrinking method to obtain the parameters of two virtual components.

The active power flexibility aggregation can be regarded as a high-dimensional polytope projection problem. In order to guarantee the disaggregation feasibility, we need to find the polytopes inscribed in the projected polytopes. We first solve the power flexibility region of virtual generator Ω^G .

Collecting all the technical constraints of GDERs, including the temporal decoupling constraints (18) and temporal coupling constraints of GDERs. They can be expressed as the compact form as

$$E^G x^{GQ} \leq f^G \quad (20)$$

Besides, based on the linear power flow model, the active output power trajectory of VG p^G can be expressed as the following affine form:

$$p^G = C^G x^{GQ} + d^G \quad (21)$$

Therefore, the origin polytope and corresponding projected polytope are:

$$\Omega_1^G = \left\{ \begin{bmatrix} x^{GQ} \\ p^G \end{bmatrix} \mid E^G x^{GQ} \leq f^G, p^G = C^G x^{GQ} + d^G \right\} \quad (22)$$

$$\Omega_2^G = \left\{ p^G \mid A^{G*} p^G \leq b^{G*} \right\} \quad (23)$$

Since solving the precise value of A^{G*} and b^{G*} is an NP-hard problem, we can use an inscribed polytope that has similar shape as Ω_2^G to approximate it. Because the cluster aggregated by GDERs still has the characteristics of generators, the power flexibility region Ω^G of virtual generator can be described by several parameters, including the ramp rates ($\bar{r}_t^G$ and $\underline{r}_t^G$) and upper and lower power bounds ($\bar{p}_t^G$ and $\underline{p}_t^G$) at each time period. That is

$$\Omega^G = \left\{ p^G \mid \begin{array}{l} \underline{r}_t^G \leq p_t^G - p_{t-1}^G \leq \bar{r}_t^G, \\ \underline{p}_t^G \leq p_t^G \leq \bar{p}_t^G, \forall t \in \mathcal{T} \end{array} \right\} \quad (24)$$

The polytope Ω^G in (24) can be further expressed as the following compact form:

$$\Omega^G = \left\{ p^G \mid A^G p^G \leq b^G \right\} \quad (25)$$

where A^G is a constant matrix as (26); b^G is a vector composed of the key parameters as (28). The exact value of b^G is obtained with the high-dimensional polytope based bound shrinking method in Section III.

$$A^G = [I_T \ -I_T \ \Lambda^\top \ -\Lambda^\top]^\top \quad (26)$$

$$\Lambda = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 & 1 \end{bmatrix}_{(T-1) \times T} \quad (27)$$

$$b^G = \left[(\bar{p}^G)^\top - (\underline{p}^G)^\top \ (\bar{r}^G)^\top - (\underline{r}^G)^\top \right]^\top \quad (28)$$

Then, we can calculate the flexibility region of VB denoted as Ω^B with similar process. Ω^B can be described with the parameters including the energy constraints ($\bar{E}_t^B$ and $\underline{E}_t^B$) and the upper and lower bounds ($\bar{p}_t^B$ and $\underline{p}_t^B$) at each time period. That is

$$\Omega^B = \left\{ p^B \mid \begin{array}{l} E_t^B = p_{t-1}^B \Delta t + E_{t-1}^B, \\ \underline{E}_t^B \leq E_t^B \leq \bar{E}_t^B, \\ \underline{p}_t^B \leq p_t^B \leq \bar{p}_t^B, \forall t \in \mathcal{T} \end{array} \right\} \quad (29)$$

The corresponding compact form of (29) is:

$$\Omega^B = \{ p^B \mid A^B p^B \leq b^B \} \quad (30)$$

where A^B is a constant matrix as (31); b^B is a vector composed of the parameters as (33).

$$A^B = [I_T \ -I_T \ \Gamma^\top \ -\Gamma^\top]^\top \quad (31)$$

$$\Gamma = \Delta t \cdot \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}_{T \times T} \quad (32)$$

$$\mathbf{b}^B = \begin{bmatrix} (\bar{\mathbf{p}}^B)^\top & -(\underline{\mathbf{p}}^B)^\top & (\bar{\mathbf{E}}_t^B)^\top & -(\mathbf{E}_t^B)^\top \end{bmatrix}^\top \quad (33)$$

Finally, the power flexibility of VPP can be expressed as follows:

$$\Omega^{\text{VPP}} = \left\{ \begin{bmatrix} \mathbf{p}^{\text{VPP}} \\ \mathbf{q}^{\text{VPP}} \end{bmatrix} \middle| \mathbf{p}^{\text{VPP}} = \mathbf{p}^B + \mathbf{p}^G, \mathbf{p}^B \in \Omega^B, \mathbf{p}^G \in \Omega^G, (p_t^{\text{VPP}}, q_t^{\text{VPP}}) \in \mathcal{R}_t, \forall t \in \mathcal{T} \right\}. \quad (34)$$

III. ESTIMATE THE PARAMETERS OF THE VG AND VB

In this section, we obtain the parameters $\mathbf{b}^G$ of VG as an example, then $\mathbf{b}^B$ of VB can be obtained with the same process.

We use an iterative method to obtain $\mathbf{b}^G$. It corresponds to an inscribed polytope Ω^G that meets $\Omega^G \subseteq \Omega_2^G$, while keeping the flexibility region Ω^G as big as possible. In the iteration process, the slopes of hyperplanes A^G remain constant, but the intercepts $\mathbf{b}^G$ change, as illustrated in (c) of part (ii) of Fig. 1. That is to say, the type of constraints remains unchanged, but the parameters are adjusted to guarantee the disaggregation feasibility. Denote the vector in the k -th iteration as $\mathbf{b}_{(k)}^G$, and the corresponding polytope is $\Omega_{(k)}^G$. We start from the circumscribed polytope $\Omega_{(0)}^G$, so that $\mathbf{b}_{(0)}^G$ can be easily obtained by solving series of linear optimization problems. For $\forall t \in \mathcal{T}$

$$\bar{p}_{t,(0)}^G = \left\{ \max p_t^G \middle| (\mathbf{x}^{\text{GQ}}, \mathbf{p}^G) \in \Omega_1^G \right\} \quad (35)$$

$$\bar{r}_{t,(0)}^G = \left\{ \max r_t^G = p_t^G - p_{t-1}^G \middle| (\mathbf{x}^{\text{GQ}}, \mathbf{p}^G) \in \Omega_1^G \right\} \quad (36)$$

Similarly, the value of $\bar{p}_{t,(0)}^G$ and $\bar{r}_{t,(0)}^G$ can be obtained by solving the minimization optimization problem. The initial parameters $\mathbf{b}_{(0)}^G$ and the power flexibility region $\Omega_{(0)}^G$ are obtained. Then, we need to gradually adjust the key parameters of $\Omega_{(k)}^G$ in the iteration process, making $\Omega_{(k)}^G$ shrink step by step. Finally, $\Omega_{(k)}^G \subseteq \Omega_2^G$ is realized for a certain $\mathbf{b}_{(k)}^G$. We use the polytope based bound shrinking method to realize this process, which is divided into two stages: infeasible point search and parameters modification.

In the first stage, we identify the infeasible point $\mathbf{p}_{(k)}^G$, that satisfies $\mathbf{p}_{(k)}^G \in \Omega_{(k)}^G$ and $\mathbf{p}_{(k)}^G \notin \Omega_2^G$, simultaneously. The infeasible extreme point $\mathbf{p}_{(k)}^G$ can be obtained by solving a Stackelberg game [17], i.e., the following min-max problem.

$$f_{(k)}(\mathbf{b}_{(k)}^G) = \min_{\mathbf{p}^G} \max_{\mathbf{x}^{\text{GQ}}} 0 \quad (37a)$$

$$\text{s.t. } A^G \mathbf{p}^G \leq \mathbf{b}_{(k)}^G \quad (37b)$$

$$E^G \mathbf{x}^{\text{GQ}} \leq \mathbf{f}^G \quad (37c)$$

$$\mathbf{p}^G = C^G \mathbf{x}^{\text{GQ}} + \mathbf{d}^G \quad (37d)$$

In this Stackelberg game, the leader is the active power trajectory of virtual generator $\mathbf{p}^G$, and the follower is $\mathbf{x}^{\text{GQ}}$. The leader wants to find a bad trajectory that is within the flexibility region (37b), but cannot be disaggregated as (37d) while keeping (37c) holds. If there exist such a trajectory $\mathbf{p}_{(k)}^G$, that cannot be successfully disaggregated by adjusting $\mathbf{x}^{\text{GQ}}$, the maximum problem of follower will become infeasible, and the optimal value of (37a) is $-\infty$. Otherwise, the optimal value of (37a) equals 0.

The min-max problem (37) is very special, and has some bad characteristics. Firstly, its objective function has no gradient, and only has two possible values, 0 or $-\infty$. Secondly, the result of optimization is up to the feasibility of the follower optimization problem, which cannot be described in the computer programming language. Therefore, we transform the min-max problem (37) into a bilinear programming problem using dual method. Then, the primal infeasible situation is equivalent to a dual unbounded situation.

$$\min_{\mathbf{p}^G, \lambda, \pi} \lambda^\top \mathbf{f}^G - \pi^\top \mathbf{d}^G + \pi^\top \mathbf{p}^G \quad (38a)$$

$$\text{s.t. } A^G \mathbf{p}^G \leq \mathbf{b}_{(k)}^G \quad (38b)$$

$$\lambda^\top E^G + \pi^\top C^G = \mathbf{0} \quad (38c)$$

$$\lambda \geq \mathbf{0} \quad (38d)$$

where λ and π are the dual variables of constraints (37c) and (37d), respectively.

There is a bilinear term $\pi^\top \mathbf{p}^G$ in (38a), making it hard to solve. Therefore, we use KKT conditions to form the equivalent optimization problem of (38) and transform the complimentary slackness condition into mix integer linear constraints as (39).

$$\min_{\mathbf{p}^G, \lambda, \pi, \omega, s} \lambda^\top \mathbf{f}^G - \pi^\top \mathbf{d}^G - \omega^\top \mathbf{b}_{(k)}^G \quad (39a)$$

$$\text{s.t. } \omega^\top A^G + \pi^\top = \mathbf{0} \quad (39b)$$

$$\lambda^\top E^G + \pi^\top C^G = \mathbf{0} \quad (39c)$$

$$\lambda \geq \mathbf{0} \quad (39d)$$

$$\mathbf{0} \leq \mathbf{b}_{(k)}^G - A^G \mathbf{p}^G \leq M(\mathbf{I} - \mathbf{s}) \quad (39e)$$

$$\mathbf{0} \leq \omega \leq M\mathbf{s} \quad (39f)$$

$$\mathbf{W}\mathbf{s} \leq \mathbf{I}_n \quad (39g)$$

$$\mathbf{I}_n^\top \mathbf{s} \geq \mathbf{T} \quad (39h)$$

$$s \in \{0, 1\}^n \quad (39i)$$

where n is the length of $\mathbf{b}_{(k)}^G$; ω denotes the dual variable vector of (38b); M is a sufficiently large constant. According to the physical meaning, (39g) and (39h) are cutting panels to accelerate the solution of (39). If $s_i = 1$, the i -th column of constraint (37b) is active. However, the upper and lower bounds of constraint (37b) cannot be active simultaneously, as limited by the cutting panel (39g), where $\mathbf{W}$ is shown in (40). Constraint (39h) indicates that the extreme point is located at the intersection of at least T hyperplanes in the T dimensional polytope $\Omega_{(k)}^G$.

$$\mathbf{W} = \begin{bmatrix} \mathbf{I}_T & \mathbf{I}_T & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{T-1} & \mathbf{I}_{T-1} \end{bmatrix} \quad (40)$$

Besides, the clustering method [11] can be used to aggregate the homogeneous DERs located in the same community first to decrease the number of DERs and lower the complexity of (39).

By solving (39), we find the worst trajectory $\mathbf{p}_{(k)}^G$ that is not disaggregation feasible, corresponding to an extreme point of polytope $\Omega_{(k)}^G$. Then, in the second stage, we modify $\mathbf{b}_{(k)}^G$ to shrink the bounds of $\Omega_{(k)}^G$ and exclude the infeasible points $\mathbf{p}_{(k)}^G$, as shown in the schematic Fig. 2. At first, the point $\mathbf{p}_{(k)}^{\text{bd}}$

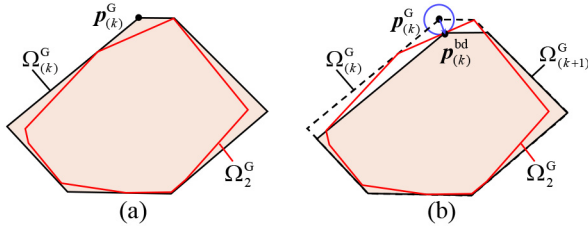


Fig. 2. Schematic diagram of bound shrinking process: (a) find the worst trajectory $p_{(k)}^G$ in the first stage, corresponding to an extreme point of polytope $\Omega_{(k)}^G$; (b) find the nearest boundary point $p_{(k)}^{bd}$ and shrink the bounds to get $\Omega_{(k+1)}^G$.

Algorithm 1 The Polytope Based Bound Shrinking Method

- 1: Initialize $k=0$, obtain the initial value $b_{(0)}^G$ by solving (35), (36) and the corresponding minimization problems
- 2: Identify the infeasible point $p_{(k)}^G$ and obtain the optimal objective value $f_{(k)}$ by solving (37)
- 3: If $f_{(k)}=0$, output $b_{(k)}^G$ and exit this algorithm, otherwise go to step 4.
- 4: Search for the boundary point $p_{(k)}^{bd}$ near to $p_{(k)}^G$ according to (41)
- 5: Update the value of $b_{(k)}^G$ to $b_{(k+1)}^G$ based on (43)
- 6: $k \leftarrow k+1$ and go back to step 2

on the boundary of Ω_2^G near $p_{(k)}^G$ can be found by solving the following optimization problem:

$$p_{(k)}^{bd} = \arg \min_{p^G} \|p_{(k)}^G - p^G\| \quad (41a)$$

$$\text{s.t. } (x^{GQ}, p^G) \in \Omega_1^G \quad (41b)$$

The solution of problem (41) is the point lying on the boundary of Ω_2^G and having the minimum distance between $p_{(k)}^G$ and Ω_2^G . Then, we identify the active constraints for $p_{(k)}^G$ in (37b) and denote the set of their row indexes as $\mathcal{I}_{(k)}^G$, that is

$$(A^G)_i p_{(k)}^G = (b^G)_i, \forall i \in \mathcal{I}_{(k)}^G \quad (42)$$

Finally, $(b^G)_{(k)}^G$ are adjusted to make $p_{(k)}^{bd}$ as the new extreme point of $\Omega_{(k+1)}^G$:

$$b_{(k+1)}^G = \arg \max_{b^G} I_n^T b^G \quad (43a)$$

$$\text{s.t. } b^G \leq b_{(k)}^G \quad (43b)$$

$$(A^G)_i p_{(k)}^{bd} \geq (b^G)_i - M(1 - z_j), \forall i \in \mathcal{I}_{(k)}^G \quad (43c)$$

$$I_J^T z \geq T, z \in \{0, 1\}^J \quad (43d)$$

where J is the cardinality of set $\mathcal{I}_{(k)}^G$. Constraint (43b) makes sure $\Omega_{(k+1)}^G \subseteq \Omega_{(k)}^G$ holds. (43c) and (43d) mean that $p_{(k)}^{bd}$ lies on at least T hyperplanes of $\Omega_{(k+1)}^G$, so that it is the extreme point of $\Omega_{(k+1)}^G$. After obtaining $b_{(k+1)}^G$, we will return to the first stage for the $(k+1)$ -th iteration.

The pseudo-code of the polytope based bound shrinking method is shown in Algorithm 1.

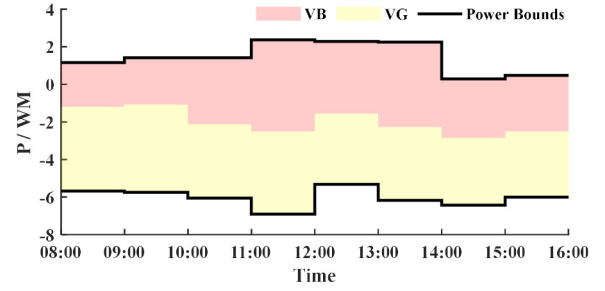


Fig. 3. The upper and lower power bounds of the aggregated flexibility.

IV. NUMERICAL TESTS

A. Simulation Setup

Numerical tests are carried out on IEEE 123 three-phase distribution network system. The DERs include 8 CHPs, 6 ESSs, 3 WTs, 8 PVs, and 12 HVACs. The detailed parameters of topology and DERs are presented in the supplementary file [25]. Historical data of available PV, WT generation, and ambient temperature curves are cited from [26].

The test cases were conducted on a laptop with Intel Core i7-8550U CPU, 1.80 GHz and 16 GB RAM. The MATLAB software with YALMIP toolbox and CPLEX solver were used to solve the optimization problems.

B. Flexibility Aggregation Results

We compare the flexibility aggregation results obtained by the existing algorithms in [5] and [17] and our method. This is because other existing algorithms cannot deal with the flexibility aggregation problem in our test case. For example, the virtual battery model [11], [12] and the Minkowski Sum based methods [13], [14] do not take the network constraints into consideration. The model in [15] does not include the energy constraints of BDERs.

In this test case, the confidence level is specified as 0.8 and the results are shown in Fig. 3. The upper and lower power bounds show that our method can exploit the widest range of the output active power. Besides, with our method, the VPP is equivalent to a VB and a VG. Accordingly, the range of output power can be separated into two parts: the pink and yellow region in Fig. 3. According to the model in (24) and (29), the VB and VG also have temporal coupling parameters, such as the ramp rates and energy constraints as shown in Fig. 4.

C. Comparison of Conservation

The existing flexibility evaluation methods include inscribed hyperboxes method [5], [17] and inscribed ellipsoid method [18]. All the methods involve graphics in high-dimensional space that are unable to draw directly. Therefore, to compare the conservation level of the existing algorithms and our method, we project the high-dimensional results onto a two-dimensional plane. They show the flexibility constraints of two adjacent time points.

According to the results in Fig. 5, our proposed method can explore the biggest flexibility region in each projected figure and has the least conservation level. This is because

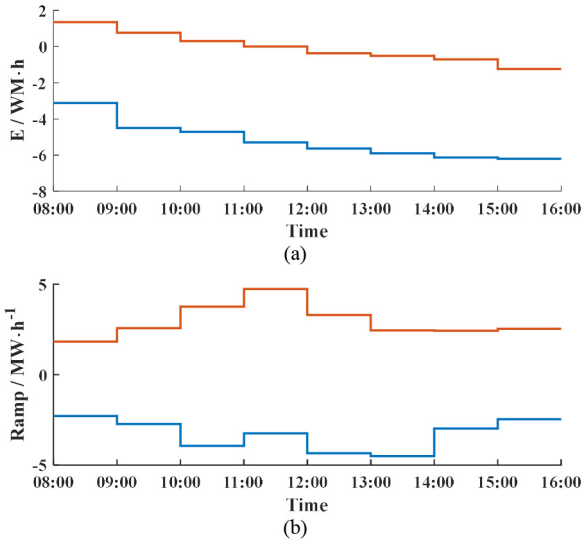


Fig. 4. The parameters of VB and VG: (a) The upper and lower energy bounds of VB; (b) The up and down ramp rate of VG.

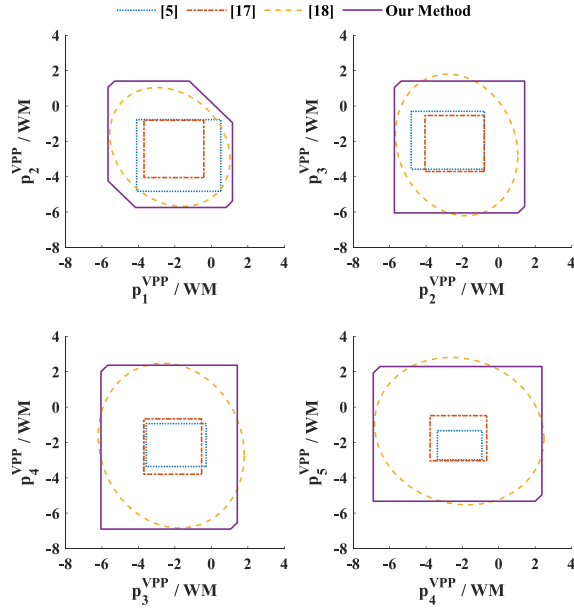


Fig. 5. Projected flexibility region approximations obtained by different methods: (a) 8:00-9:00; (b) 9:00-10:00; (c) 10:00-11:00; (d) 11:00-12:00.

our proposed method classifies different types DERs into GDERs and BDERs, and characterizes their flexibility as VG and VB, respectively. The shapes of the flexibility polytopes Ω^G and Ω^B we obtained are similar to the shapes of projected polytopes Ω_2^G and Ω_2^B , thus having better approximation performance.

What is more, scenario coverage rate method is used to test the conservation level of different methods. We generate 100,000 random trajectories that can actually be successfully disaggregated by the DERs in VPP. Since the evaluation result of flexibility is obtained by approximation, there must be some trajectories that are mistakenly considered infeasible. By testing the scenario coverage rate of the feasible trajectories, we can analyse the conservation level of different methods.

TABLE I
COMPARISON OF CONSERVATION

The ratio of BDERs to total capacity		20%	40%	60%	80%
Scenario coverage rate	[5]	67.90%	61.38%	47.28%	39.30%
	[17]	67.64%	62.16%	52.22%	49.46%
	[18]	79.15%	80.33%	77.29%	76.78%
	Our method	82.52%	84.15%	83.54%	85.70%

With different capacity rate of the ESS and HVAC devices to the total capacity of all the DERs in this test case, the conservation of different algorithms are tested, as shown in Table I.

Because the ramp rates and energy constraints of heterogeneous DERs are considered explicitly, the complementary characteristic of active power in different time periods are fully exploited. The conservation of our method does not vary too much as the ratio of BDERs to total capacity increases. Therefore, our proposed method can obtain a higher scenario coverage rate than the existing methods, especially in the setting with high ratio of BDERs to total capacity.

V. CONCLUSION

In this paper, we proposed a novel method to evaluate the aggregated power flexibility of virtual power plant. The origin problem involves numerous decision variables, and the network constraints of VPP and temporal coupling constraints of DERs will make all the variables interdependent. Our proposed method makes the problem trackable by

- 1) modeling the power flexibility aggregation evaluation process as a projection problem of high-dimensional polytope;
- 2) using a virtual generator and a virtual battery to represent the power flexibility of original VPP; and
- 3) calculating the parameters of equivalent virtual generator and battery with a polytope based bound shrinking method.

The results of numerical test verified the disaggregation feasibility of the evaluation results, and showed the lower conservation than the existing methods.

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